

An Investigation into Models for Facial Expression Classification

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1. Introduction

Facial expression classification is a challenging computer vision problem where the spatial arrangement of facial features, such as eyebrows, eyes, cheeks, and mouth, carries most of the predictive signal. Different expressions are characterized by distinct, localized visual patterns. For example, happiness often involves an upturned mouth and raised cheeks while surprise tends to be displayed with widened eyes and lifted eyebrows. Each family of machine learning models captures patterns of images differently, so we chose a set of models that progressively incorporate more of the structure and non-linear patterns present in images.

Our driving question was to see how much explicitly modeling spatial structure improved generalization in facial expression recognition. To answer this we decided to compare three model families of increasing spatial awareness: (1) an RBF-kernel SVM after PCA as a strong baseline, (2) a MLP to test whether nonlinear learning is sufficient when the model still lacks locality, and (3) a CNN designed to learn increasingly higher-level facial cues along with a ResNet based transfer learning model which is designed to classify image categories. This progression helps us justify whether increased model complexity translates to improved test performance and better generalization of facial expressions.

Our MLP can achieve an overall accuracy of 44.82%, which performs worse than our SVM with an accuracy of 45.81%. This result in our MLP may be due to overfitting on the training set or only being able to capture more significant facial expression patterns, like happiness or surprise. Our CNN dramatically improved upon these two models and can achieve overall accuracy of 64.73%. Our ResNet Transfer Learning Model achieved an overall test accuracy of 48.71%, but training accuracy of 67.04%.

2. Facial Expression Dataset

For this project, we used the FER2013 dataset collected by Pierre-Luc Carrier and Aaron Courville as part of a research project at the University of Montreal. This dataset initially contained 48x48 resolution 35,887 B/W images across 7 emotions (see Figure 1), but was reduced down to 35,340 images after filtering out disgust-labeled images, leaving us with 6 emotions to classify. We did this because the number of “disgust” images was small in

proportion to the other classes of the dataset, changing our minority class from having 547 images to 4002 images. This would result in more practical balancing strategies and still have enough data to learn patterns in the facial expression images. See Table 1 for the full class distribution.

Class Distribution	
	Count
Happy	8989
Neutral	6198
Sad	6077
Fear	5121
Angry	4953
Surprise	4002
Disgust	547

Table 1: Class distribution before data filtering and augmentation

To account for the imbalance in the data for SVM and CNN, undersampling was used to train the model on a balanced subset of the set. For the SVM specifically, PCA dimensionality reduction was performed to reduce the flattened 2304-dimensional feature space to about a 250-dimensional representation (almost 11% of the initial feature space), while still retaining 95% of the variance.

Data Augmentation was performed on the dataset to raise the amount of training pictures. Flipping the images horizontally, and slightly rotating the images resulted in 84,816 training images and 7,068 test images.

A small subset of images in the dataset were either not recognizable human faces, instead appearing as pixel artifacts, or highly cartoon representations of people. While the majority of samples consisted of real human faces, these anomalous images were identified late in the project timeline, leaving insufficient time for systematic filtering. Because of this, these samples were left in the dataset. We assume that their low frequency limits overall impact on model training and evaluation.



Figure 1: Example images of six facial expressions included in the dataset (angry, fear, happy, neutral, sad, and surprise respectively)

3. Baseline SVM

As a baseline for comparison, we implemented a support vector machine (SVM) algorithm with an RBF-kernel. In doing this, we chose to undersample (without disgust images) to balance the model's training dataset, leaving us with 14,412 images to work with—a manageable training size for SVM. We then fit PCA on our training data in order to reduce dimensionality, afterwards transforming validation and test sets for evaluating the model. This would allow us to capture non-linear patterns while still keeping 95% of the variance after PCA, going from 2304 features down to 254 features. This also allowed SVM to model nonlinear decision boundaries with lower computational cost and reduced risk of overfitting in a high-dimensional space. Lastly, we performed hyperparameter optimization on a validation set to select values for C and gamma, which adjusted the strength of regularization and the width of the RBF decision boundary with the goal of maximizing the model's validation F1-macro.

4. Multi-Layer Perceptron (Neural Network)

A three layer feed-forward fully connected neural network was implemented as another multi-classification model. This network had an input layer of flattened 2,304-dimensional vectors, two hidden layers with 128 and 64 neurons, and an output layer with 6 neurons. No activation function is applied to the output layer, as raw logits are required for the cross-entropy loss function. A diagram of the neural network structure on Figure 2.

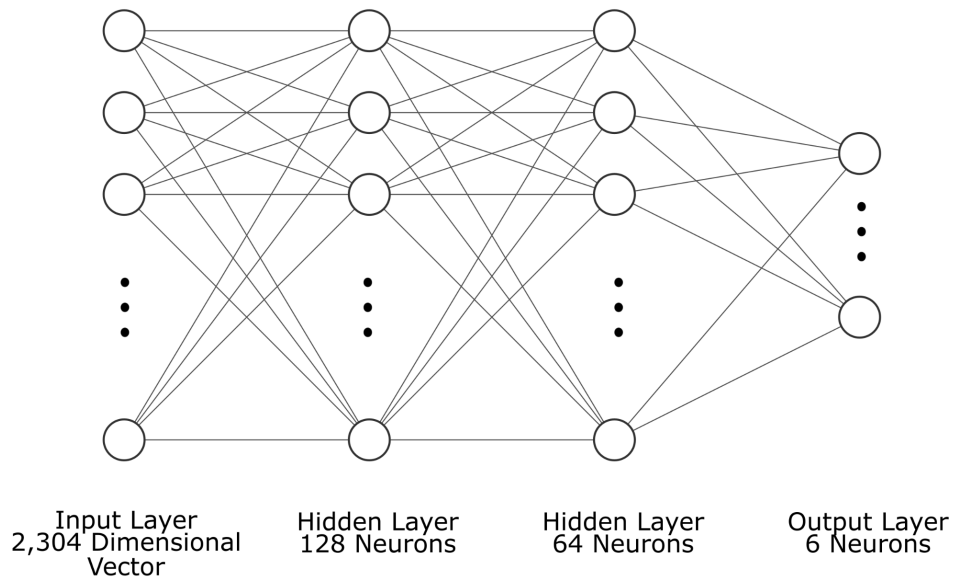


Figure 2: Diagram of the 3-layer feed-forward full-connected neural network.

Between each layer, ReLU activations were applied as a nonlinearity and then a dropout regularization ($p=0.3$) to prevent overfitting. This model was trained on 30 epochs with batch size of 512, using the Adam optimizer with a learning rate of 0.001 and L2 weight decay of $1e-4$ for regularization. Training and evaluation were performed using a standard supervised learning loop, with performance metrics logged at regular intervals. Loss and accuracy trends were tracked over time to monitor convergence and potential overfitting.

Hyperparameters, including learning rate, batch size, and dropout probabilities, were selected empirically based on experiments and common best practices for MLPs on image-based datasets. For example: dropout layers were added after each hidden layer to mitigate overfitting, the batch size was chosen to balance computational efficiency and gradient stability, the learning rate was selected to ensure stable convergence. Configurations were iteratively refined based on model performance and training.

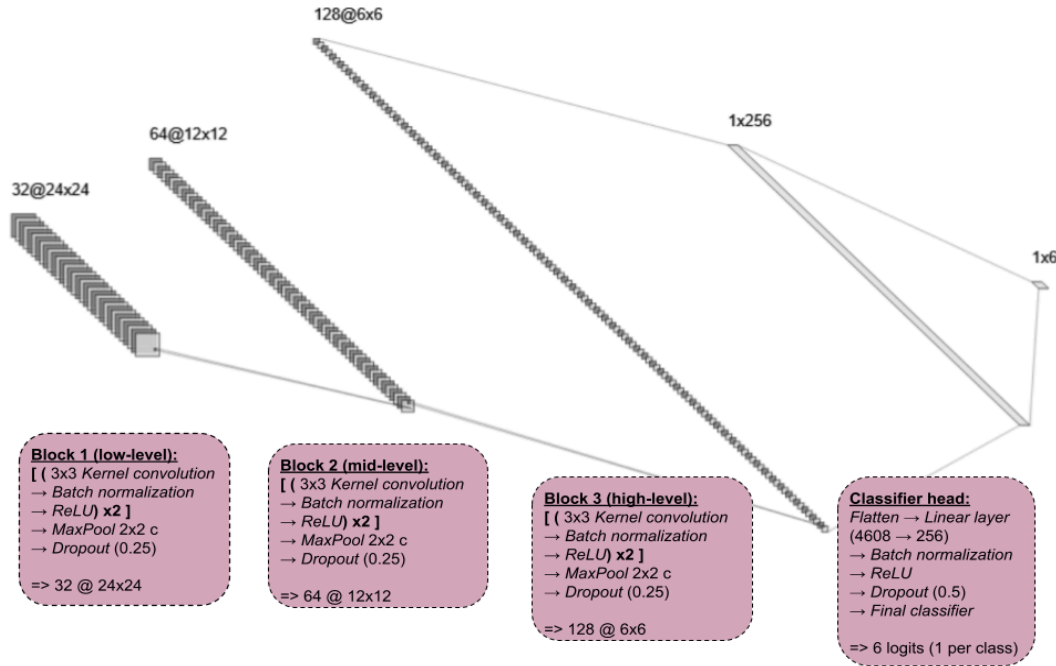


Figure 3: Visualization of convolutional neural network architecture

5. Convolutional Neural Network

To better capture the spatial structure in facial images, we implemented a convolutional neural network (CNN) for multi-class facial expression classification. The network uses three convolutional blocks, each consisting of 3x3 convolutions with batch normalization and ReLU to stabilize training and introduce non-linearity. This was then followed by 2x2 max pooling and dropout for downsampling and regularization. Block depths increased from 32 to 64 to 128,

allowing the model to go from learning low-level edges/textures to larger facial structures and emotion-specific features. The final 128x6x6 feature map is flattened and passed through the classification layer, consisting of 256 units with dropout and a linear layer to produce class logits for each emotion class. The model was trained using the Adam optimizer, with early stopping implemented to prevent overfitting, with training stopping after 16 epochs.

6. ResNet Transfer Learning Model

Using models pre-trained on large datasets with extensive computational resources can often improve performance. Our model uses a pre-trained ResNet-18 backbone. The original ImageNet classification is replaced with a custom three-layer classifier consisting of fully connected layers with batch normalization and dropout for regularization. The final layer4 is also unfrozen to perform some fine-tuning. The frozen lower layers preserve general visual features learned from ImageNet, while the fine-tuned upper layers and classification adapt to specific patterns in the FER2013 dataset.

The model uses a cross entropy loss function with label smoothing, an Adam optimizer with weight decay of 0.01, and learning rate of 0.0001 for the fine-tuning layer, and learning rate 0.001 for the classification layer. The model also performed early stopping, stopping when 3 epochs in a row showed an increase in testing loss, which resulted in a total of 7 epochs when training. A batch size of 64 was used to balance computational efficiency and gradient stability.

7. Results

To evaluate the success of all models explored, we report overall accuracy, accuracy per class, F1-score per class, and AUC in a one-vs-rest pattern. Lastly, the confusion matrix of each model will be evaluated to see how differentiable the classes are. All metrics are reported on the test set.

7.1 SVM

Using SVM with dimensionality reduction from PCA, we achieve an overall test accuracy of 45.81% and macro F1-score of 0.45. Given SVM's limitations on such a task, this approach was able to classify happy and surprised images surprisingly well, achieving class-wise accuracies of ~60% and ~66% respectively, and F1-scores of 0.59 and 0.63. Neutral was predicted at a moderate level, with an accuracy and F1-score of 0.42 for both. However, for the remainder of classes, SVM performed much worse, suggesting that the SVM was able to capture stronger, more distinct expressions, but struggled with subtler patterns of facial expressions.

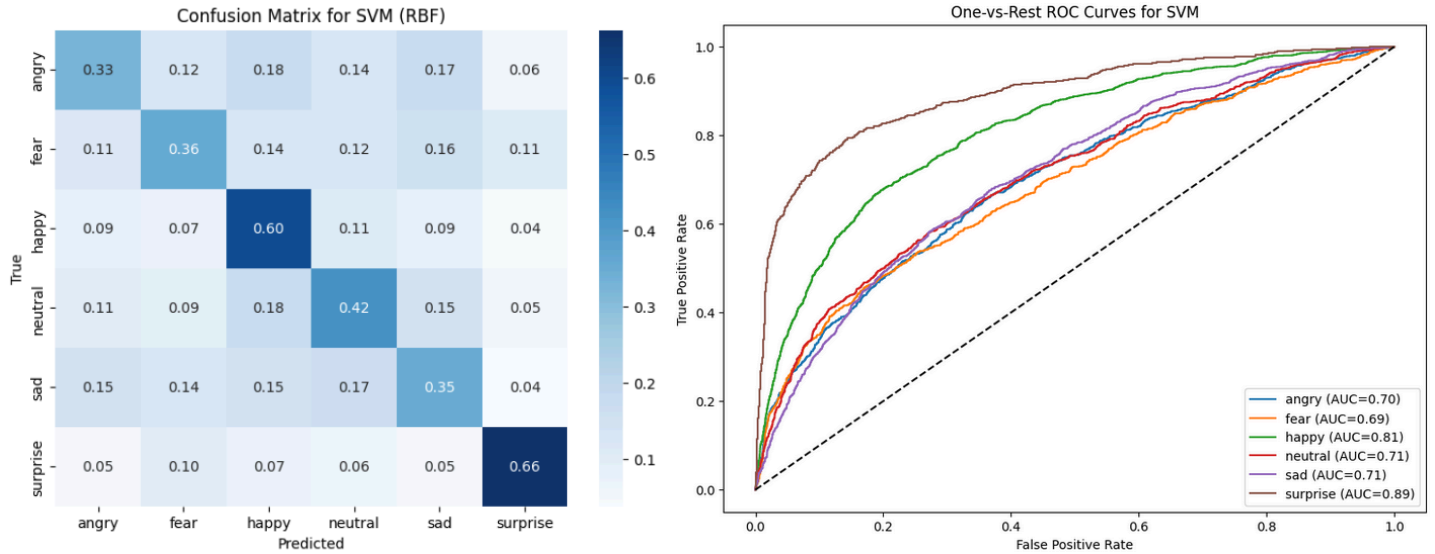


Figure 4: Confusion matrix and ROC curves of SVM

The confusion matrix and ROC curves in Figure 4 highlight several systematic errors. While surprise and happy were often correctly identified, neutral images were frequently classified as happy or sad faces, indicating difficulty in modeling subtle facial cues that might distinguish a neutral expression from similar expressions. Misclassifications of angry, fear, and sad faces were more prevalent, showing yet another limitation of this model in its ability to distinguish visual features. The ROC curves further reinforce this difficulty, showing stronger separability for happy/surprise and weaker separability for the more confusable classes.

7.2 MLP

The neural network achieved an overall accuracy of 44.8%, with the highest per-class accuracy being 59.9% for surprise, and lowest per-class accuracy being 24.6% for fear. The macro AUC score achieved was 0.77, and macro F1 score 0.44. A One vs Rest ROC Curve can be seen at Figure 5.

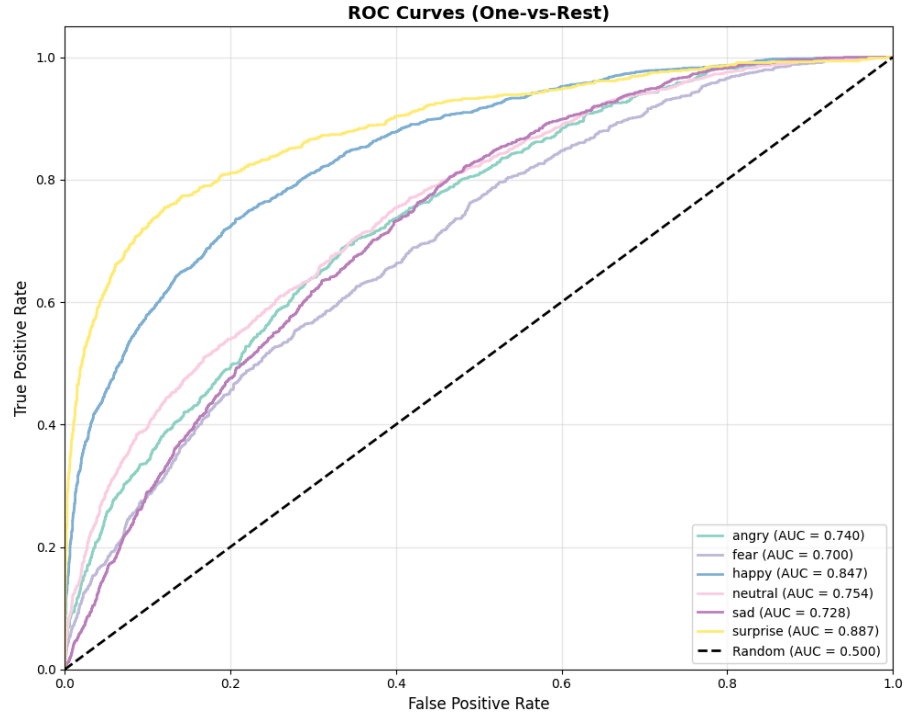
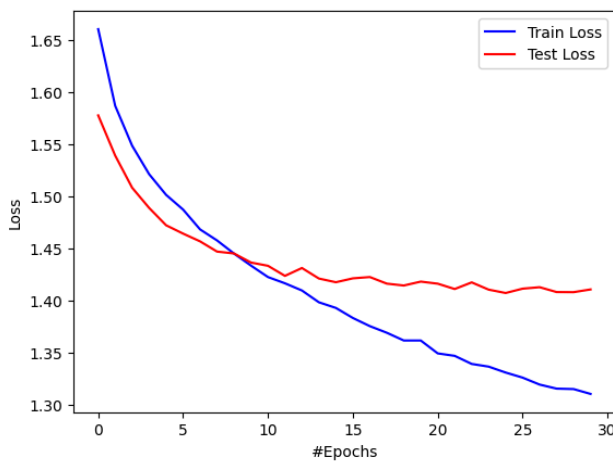
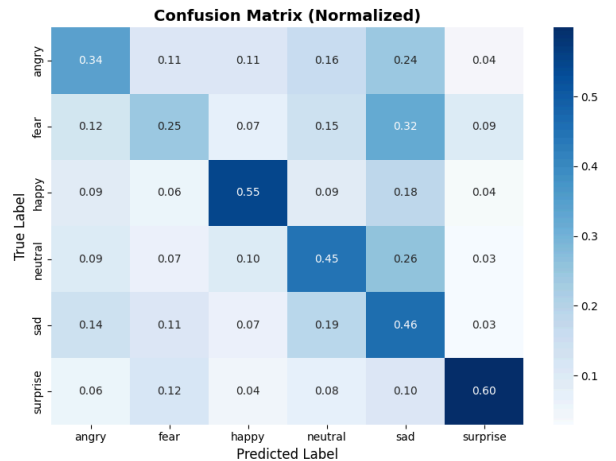


Figure 5: One vs Rest ROC Curve for MLP

The train test loss curve can be seen on Figure 6a, which shows that the model is not overfitting and stops at a good point since the test loss is flattening out, and the train loss is starting to flatten out. The neural network confuses fear the most with sadness, classifying correctly 25% of the time while classifying incorrectly as sad 32% of the time. The neural network also confuses angry, neutral, and sad often. The model performed best on happiness and surprise. The confusion matrix can be seen on Figure 6b.



(a) Train test loss for 30 epochs of training with batch size 512



(b) Normalized confusion matrix on the test set

Figure 6: MLP Training Results

The model most often misclassifies emotions as sad, and 55.18% of images are misclassified. Figure 7 displays images the model correctly identified and is most confident in (left) and least confident in (right).



Figure 7: MLP prediction confidence with dataset images

7.3 CNN

Our convolutional neural network was able to produce the strongest overall performance of all models, achieving a test accuracy of 64.7% and macro F1-score of 0.62. Performance peaked in classes with strong, distinct facial cues, classifying happy particularly well with an F1 of 0.85 and accuracy of 0.87, as well as surprise with an F1 of 0.75 and accuracy of 0.84. The model also showed improved performance on neutral faces, which is a notable improvement since neutral expressions are often very subtle and can be easily confused with other expressions.

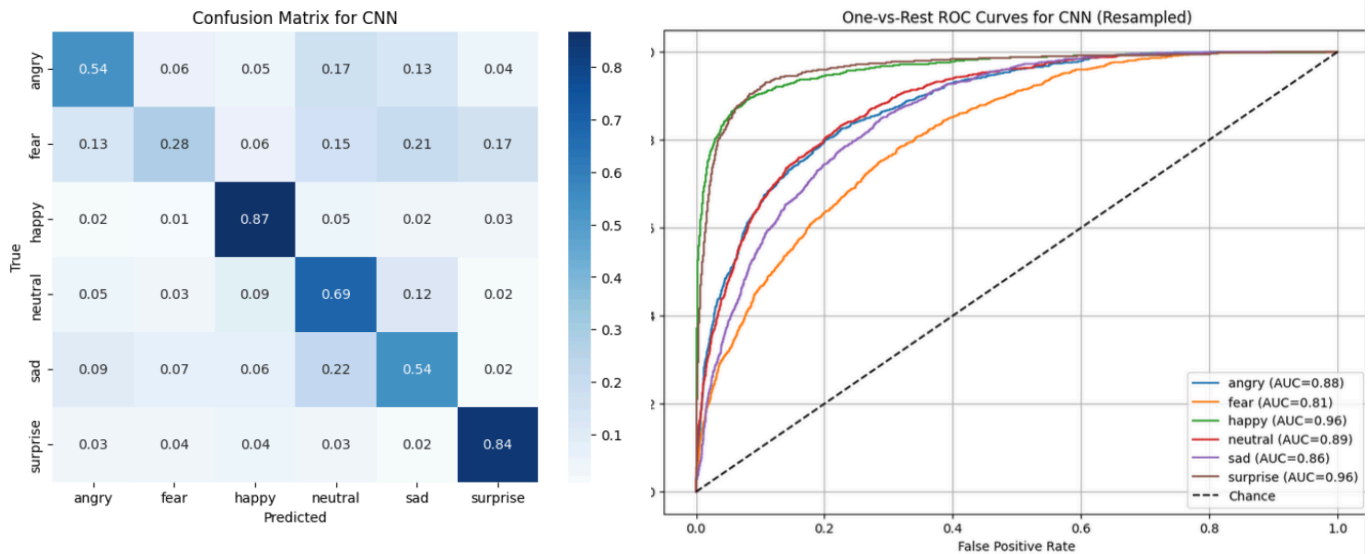


Figure 8: Confusion matrix and ROC curves of CNN

However, the CNN still struggled on fear, with a recall of 0.28 and F1 of 0.38, indicating that many fear images were misclassified as other emotions. This aligns with the difficulty of

separating fear from similar expressions (often neutral, surprise, or sadness), suggesting that fear is the primary limiting class for the overall performance of the CNN. However, angry and sad faces performed pretty moderately, achieving F1 scores of 0.56 and 0.54 respectively. This indicates that the CNN learned useful patterns, but still shows limitations between more negative expressions.



Figure 9: 16 images classified by the CNN (8 correct, 8 incorrect)

In terms of training cost, the CNN required significantly more time for training than classical models, taking approximately 40 minutes for 16 epochs. Despite this longer runtime, the gain in balanced performance and overall accuracy makes the CNN the most effective approach among the models we assessed.

7.4 ResNet Transfer Learning Model

The transfer learning model performed better than the MLP but worse than the CNN. The model has a train accuracy of 67.04% and test accuracy of 48.71%. The train test curve can be seen on Figure 10 (left), which shows the model stopping early at epoch 7 from the test loss flattening out and slightly starting to increase. The confusion matrix can also be seen on Figure 10 (right)

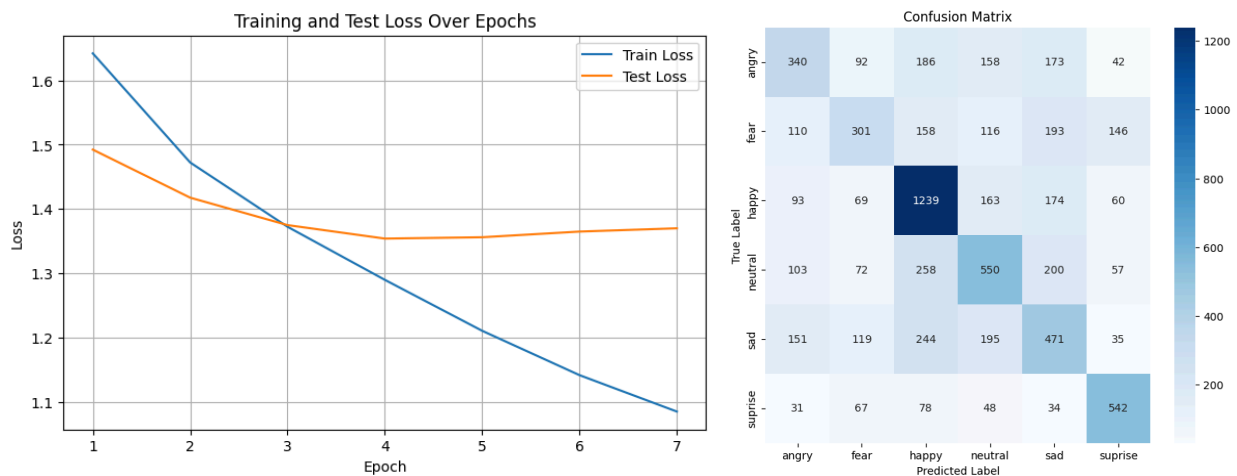


Figure 10: ResNet-18 transfer learning model training results

The model is good at identifying happy and surprising emotions, but does not perform as well on emotions like fear and sadness. The model outperformed the MLP on classifying all 6 emotion classes, but underperformed against the CNN. This could be because CNN is better trained for identifying emotions, while ResNet is trained on classifying image content. The ResNet Transfer Learning model has overall AUC score 0.8, with individual class AUC and ROC curve seen on Figure 11 (left). The models overall F1-Score is 0.47, with the best F1-Score being 0.64 and worst being 0.35. Each class's precision, recall, and F1-score can be seen on Figure 11 (right). The ResNet Transfer Learning model has a worse F1-Score and AUC score than the CNN, which suggests that ResNet does not transfer well to emotion classification.

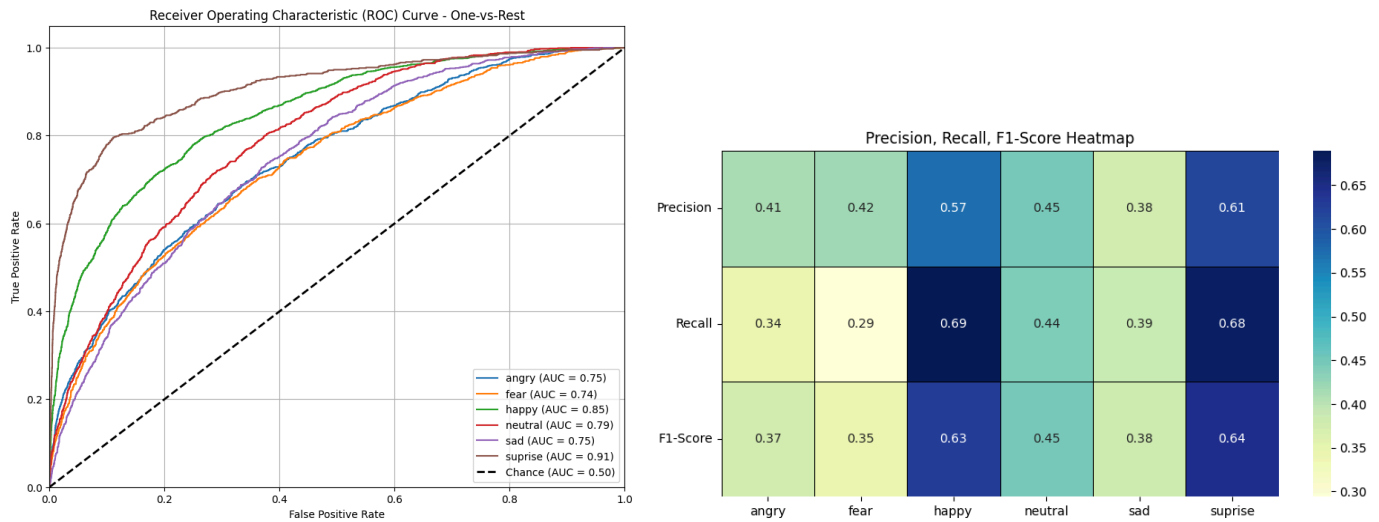


Figure 11: ResNet-18 transfer learning model numerical breakdown

The model when very confident with its predictions gets it correct most of the time, and when getting incorrect predictions has less confidence. Some incorrect predictions have unclear facial emotions that could have been mislabeled in the dataset, as seen in Figure 12 (top left).

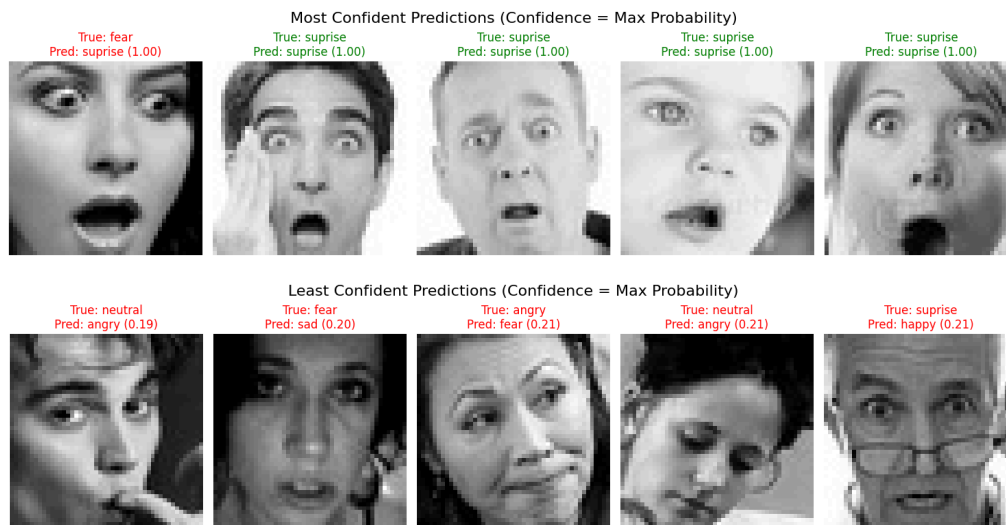


Figure 12: 5 correctly (top) and 5 incorrectly (bottom) classified images with confidence score

8. Discussion

Our results show that model performance improves when the model is capable of better preserving and learning the spatial structure of facial images. The SVM provided a useful and interpretable baseline, but was limited by its inability to learn from localized facial cues. This appeared clearly in the confusion matrix, with visually distinctive expressions being more easily separable, while neutral and confusable negative emotions were frequently mixed. The MLP performed similarly to the SVM overall, which is consistent with their key limitations of missing the structure encoded by smaller, more local regions of pixels. It showed a strong tendency of misclassifying multiple emotions as sad, showing that the model learned a separation of positive vs negative emotions, but struggled to find reliable boundaries between more confusable expressions of emotions.

In contrast, the CNN substantially improved both overall accuracy and macro F1, supporting our initial motivation that convolution is well-suited for facial expression recognition. It was able to learn more robust patterns for all emotions—particularly happy and surprised emotions, reaching 87% and 84% respectively. Negative emotions like fear were much more limiting, showing that the expressions may have overlapped with sad or surprised faces. The ResNet Transfer Learning model also improved overall accuracy, F1 score, and AUC score over the SVM and MLP models, further supporting that convolution is better suited for facial expression recognition. The ResNet model underperformed compared to the CNN, which suggest that identifying smaller finer facial features is important when it comes to emotion classification, something that the ResNet struggled to do because of its overgeneralization from pre-training.

Several limitations may have influenced the results in this analysis. First, the dataset contained anomalous and noisy images—including non-photographic faces from animated shows or some of just random clusters of pixels. Even watermarks (e.g. “shutterstock”) can add noise and reduce achievable model performance, especially for minority or ambiguous classes. Second, the use of undersampling for SVM and CNN models helps mitigate the class imbalance, but reduces the diversity of the training set, which can limit generalization. Finally, some emotion labels can be subjective and sometimes inconsistent in the dataset, meaning that some misclassifications may reflect ambiguous interpretations of expressions rather than model failure.

A way this could be dealt with in future explorations would be more intensive data cleaning to remove corrupted/anomalous images and replacing undersampling with class-weighted loss so the model can train on the full dataset while still addressing imbalance concerns. These changes should improve robustness and reduce confusion among the most similar emotion categories.

GitHub repository: <https://github.com/owenp33/facial-expression-classification>